

1. The dimensions of energy are:  
 (a)  $[M^1L^{-1}T^{-1}]$   
 (b)  $[M^2L^2T^2]$   
 (c)  $[M^1L^{-2}T^{-1}]$   
 (d)  $[M^1L^2T^{-2}]$  [d]
2. Which one of the following statements is incorrect about electromagnetic waves?  
 (a) Electromagnetic waves are deflected by an electric field and magnetic field  
 (b) They can also move in vacuum  
 (c) They have electric and magnetic components that are mutually perpendicular  
 (d) They move with a speed of  $3 \times 10^8$  m/s [a]
3. If three resistors of equal resistance  $R$  is in parallel and this combination is in series with another resistance  $R$ , then the equivalent resistance across the circuit will be \_\_\_\_\_  
 (a)  $4R / 3$   
 (b)  $3R / 4$   
 (c)  $R / 3$   
 (d)  $2R / 3$  [a]
4. In an AC circuit  $V = 100 \sin(100t)$  volts and  $I = 100 \sin(100t + \frac{\pi}{3})$  mA. The power dissipated in the circuit is  
 (a)  $10^4$  W  
 (b) 2.5 W  
 (c) 10 W  
 (d) 5.0 W [b]
5. Torque depends on  
 (a) Mass of the object  
 (b) Radius of the object's path and force  
 (c) Force and moment arm  
 (d) Mass and moment arm [c]
6. A light ray is passing through a thin prism of prism angle  $8^\circ$ . Find the refractive index of the prism if the angle of deviation is  $2^\circ$ .  
 (a) 5  
 (b) 2  
 (c) 1.25  
 (d) 2.25 [c]
7. When the amount of work done is 333 cal and change in internal energy is 167 cal, then the heat supplied is  
 (a) 166 cal  
 (b) 333 cal  
 (c) 500 cal  
 (d) 400 cal [c]
8. Kepler's law of "Areal Velocity is constant" is equivalent to law of conservation of  
 (a) Mass  
 (b) Energy  
 (c) Linear Momentum  
 (d) Angular Momentum [d]
9. The efficiency of Carnot heat engine is 75% and the amount of heat absorbed is 300 Joule, then the work done is?  
 (a) 225 J  
 (b) 300 J  
 (c) 220 J  
 (d) None of these [a]
10. In the isothermal process if the heat given to the system is 100J, the change in internal energy and the work done of the system will be  
 (a)  $W = 100J$  and  $\Delta U = 0J$   
 (b)  $W = 0J$  and  $\Delta U = 100J$   
 (c)  $W = 50J$  and  $\Delta U = 50J$   
 (d) None of these [a]
11. A body is falling freely in a viscous liquid. If the radius of the body is doubled, its terminal velocity will become:  
 (a) Doubled (b) Four times  
 (c) One fourth (d) Half [b]
12. The soap bubble has radii in the ratio 5 : 3. What is the ratio of excess pressure inside them?  
 (a) 3 : 5  
 (b) 5 : 3  
 (c) 5 : 2  
 (d) 2 : 5 [a]
13. A charge at rest produces:  
 (a) Magnetic field only  
 (b) Electric field only  
 (c) Both electric field and magnetic field  
 (d) Neither electric field nor magnetic field [b]
14. When a light ray travels from one medium to another then which of the following property remains same?  
 (a) Velocity of light  
 (b) Wavelength of light  
 (c) Frequency of light  
 (d) All of the above [c]
15. In which of the following statements, the obtained impure semiconductor is of p-type?  
 (a) Germanium is doped with bismuth  
 (b) Silicon is doped with antimony  
 (c) Germanium is doped with gallium  
 (d) Silicon is doped with phosphorus [c]
16. The condition for total internal refraction  
 (a) light must travel from denser to rarer medium  
 (b) light must travel from rarer to denser medium  
 (c) angle of incidence is lesser than critical angle  
 (d) angle of incidence is equal to the critical angle [a]

17. The efficiency of carnot engine operating between  $127^{\circ}\text{C}$  and  $327^{\circ}\text{C}$  is  
(a) 25%  
(b) 30%  
(c) 33.3%  
(d) 75% [c]
18. As we go from poles to equator the value of  $g$  \_\_\_\_\_.  
(a) Increases  
(b) Decreases  
(c) Remains same  
(d) Doubled [b]
19. A Force  $F = 3ax^2 - 3$  is acting on a body in the direction of force moving along X axis. Find the work done while the body displaces from  $x = 0$  to  $x = 2$ .  
(a)  $3a-3$   
(b)  $8a$   
(c)  $8a-6$   
(d)  $3a-8$  [c]
20. Consider the following statements about the light year:  
(a) It is a unit to measure astronomical time.  
(b) Light year is a unit of distance.  
(c) A light-year is a distance that light travels in a vacuum in one Julian year.  
Which of the statements given above is/are correct?  
(a) 1, 2 and 3  
(b) 1 and 2 only  
(c) 2 and 3 only  
(d) 1 only [c]
21. The power in AC circuit is given by:  
 $P = E_{\text{rms}} I_{\text{rms}} \cos\phi$   
The value of power factor  $\cos\phi$  in series LCR circuit at resonance is -  
(a) 0  
(b) 1  
(c)  $E_{\text{rms}} I_{\text{rms}}$   
(d)  $\frac{E_{\text{rms}} I_{\text{rms}}}{\sqrt{2}}$  [b]  
Correct Option - 2
22. A transformer is employed to reduce 220 V to 11 V. The primary draws a current of 5 A and the secondary 90 A. The efficiency of the transformer is  
(a) 20%  
(b) 40%  
(c) 70%  
(d) 90% [d]
23. According to photoelectric effect a photoelectron is emitted when the incident light is more than a certain minimum \_\_\_\_\_.  
(a) Frequency  
(b) Power  
(c) Wavelength  
(d) Intensity [a]
24. When the force retards the motion of a body, the work done by the force during retardation is:  
(a) zero  
(b) negative  
(c) positive  
(d) positive or negative depending upon the magnitude of force and displacement [b]
25. For which of the following temperatures of source and sink, the efficiency of a Carnot engine will be maximum?  
(a) 40 K, 20 K  
(b) 60 K, 40 K  
(c) 80 K, 60 K  
(d) 100 K, 80 K [a]
26. If  $3^x + 3^y = 3^{x+y}$  then what is  $\frac{dy}{dx}$  equal to ?  
(a)  $\frac{3^{x+y}-3^x}{3^y}$   
(b)  $\frac{3^{x-y}(3^y-1)}{1-3^x}$   
(c)  $\frac{3^x+3^y}{3^x-3^y}$   
(d)  $\frac{3^x+3^y}{1+3^{x+y}}$  [b]
27. Find  $\frac{d^2y}{dx^2}$  if  $y = \sin(\log x)$ ?  
(a)  $\frac{\sin(\log x) + x \cdot \cos(\log x)}{x^2}$   
(b)  $-\frac{\sin(\log x) + \cos(\log x)}{x^2}$   
(c)  $\frac{x \cdot \sin(\log x) - \cos(\log x)}{2x}$   
(d)  $\frac{\sin(\log x) + \cos(\log x)}{2x}$  [b]
28. If  $5 \tan A = 3$ , then the value of  $\frac{5 \sin A - 2 \cos A}{\sin A + 2 \cos A}$  is  
(a)  $7/9$   
(b)  $5/13$   
(c)  $9/13$   
(d)  $5/7$  [b]
29. Find the number of permutations of the letters of the word ALLAHABAD.  
(a) 6750  
(b) 7560  
(c) 5670  
(d) 6050 [b]
30. Three fair dice are thrown. What is the probability of getting a total greater than or equal to 15?  
(a)  $\frac{19}{216}$   
(b)  $\frac{1}{12}$   
(c)  $\frac{17}{216}$   
(d)  $\frac{5}{54}$  [d]

31. How many terms of GP 2, 4, 8, 16.....are required to give sum 254

- (a) 4  
(b) 5  
(c) 6  
(d) 7

[d]

32. If  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$  then  $\text{adj}(A^T) - (\text{adj } A)^T$  is

- (a)  $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$   
(b)  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$   
(c)  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(d)  $2(ad - bc) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

[a]

33. Find the length of the tangent to circle  $x^2 + y^2 = 169$  from the point (12, -6)?

- (a)  $\sqrt{10}$   
(b)  $\sqrt{11}$   
(c)  $2\sqrt{3}$   
(d) None of these

[b]

34. Find the direction cosines of the vector  $\hat{i} + 2\hat{j} - \hat{k}$ .

- (a)  $\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}$   
(b)  $\frac{1}{\sqrt{4}}, \frac{2}{\sqrt{4}}, \frac{-1}{\sqrt{4}}$   
(c)  $\frac{-1}{\sqrt{4}}, \frac{-2}{\sqrt{4}}, \frac{1}{\sqrt{4}}$

(d) None of these.

[a]

35. What is the value of  $\int_0^{\pi/4} (\tan^3 x + \tan x) dx$ ?

- (a)  $\frac{1}{4}$   
(b)  $\frac{1}{2}$   
(c) 1  
(d) 2

[b]

36. The value of  $\frac{\sin 72^\circ}{\cos 18^\circ}$  is :

- (a) -1  
(b) 1  
(c) 2  
(d) 0

[b]

37. The value of K which makes the function

defined by  $f(x) = \begin{cases} \sin \frac{1}{x}, & \text{if } x \neq 0 \\ k, & \text{if } x = 0 \end{cases}$

Continuous at  $x = 0$  is

- (a) 8  
(b) 1  
(c) -1

(d) None of the above

[d]

38. Find the distance between foci of the ellipse

$$\frac{x^2}{100} + \frac{y^2}{64} = 1.$$

- (a) 2  
(b) 3  
(c) 4  
(d) 12

[d]

39.  $\lim_{x \rightarrow 0} \frac{e^x - 1 - x - x^2}{x^2} =$

- (a) 0  
(b)  $\frac{1}{2}$   
(c)  $\frac{-1}{2}$   
(d) -1

[c]

40. If  $A = \{1, 2, 3\}$   $B = \{3, 4, 5\}$ , then  $(A \cap B) \times A$  is

- (a)  $\{(1, 3), (2, 3), (3, 3)\}$   
(b)  $\{(3, 1), (3, 2), (3, 3)\}$   
(c)  $\{(1, 3), (3, 1), (3, 2)\}$   
(d) none of these

[b]

41. Find the 4<sup>th</sup> term from the end in the expansion  $\left(\frac{x}{2} - x\right)^{15}$

- (a)  $\frac{455x^{15}}{8}$   
(b)  $\frac{45x^5}{8}$   
(c)  $-\frac{425x^{15}}{8}$

(d) More than one of the above

[a]

42. The median of the following data is:

31, 37, 43, 42, 25, 46, 45, 39, 32

- (a) 25  
(b) 42  
(c) 46  
(d) 39

[d]

43. The range of  $\tan^{-1} x$  is

- (a)  $\left(\pi, \frac{\pi}{2}\right)$   
(b)  $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$   
(c)  $(-\pi, \pi)$   
(d)  $(0, \pi)$

[b]

44. Find the modulus of  $(5 + \sqrt{-11})(1 + \sqrt{-5})$ .

- (a) 3  
(b)  $2\sqrt{3}$   
(c) 6  
(d)  $6\sqrt{6}$

[d]

45. If  $A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}$  and  $B = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$ , then  $A + B$ ?

- (a)  $\begin{bmatrix} -2 & 0 \\ 2 & 4 \end{bmatrix}$  (b)  $\begin{bmatrix} 0 & 4 \\ 4 & 4 \end{bmatrix}$   
(c)  $\begin{bmatrix} -1 & 4 \\ 3 & 0 \end{bmatrix}$  (d)  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

[b]

46. A relation defined as  $R = \{(1, 2), (3, 4), (2, 4)\}$  find  $R^{-1}$ .  
 (a)  $\{\emptyset, (2, 1), (4, 3), (4, 2)\}$   
 (b)  $\{\{0\}, (2, 1), (4, 3), (4, 2)\}$   
 (c)  $\{(2, 1), (4, 3), (4, 2)\}$   
 (d) None of these [c]
47. If  $1/4, 1/x, 1/10$  are in HP, then what is the value of  $x$ ?  
 (a) 5  
 (b) 6  
 (c) 7  
 (d) 8 [c]
48. If  $y = 5 \cos x - 3 \sin x$ , then  $\frac{d^2y}{dx^2} + y$  equals  
 (a)  $8 \sin x \cos x$   
 (b)  $3 \sin x \cos x$   
 (c) 1  
 (d) 0 [d]
49. The area of the region bounded by the curve  $y = x^2$  and the line  $y = 16$   
 (a)  $\frac{32}{3}$   
 (b)  $\frac{256}{3}$   
 (c)  $\frac{64}{3}$   
 (d)  $\frac{128}{3}$  [b]
50. General solution of differential equation  $\frac{dy}{dx} = \frac{y}{x}$  is:  
 (a)  $\log y = kx$   
 (b)  $y = kx$   
 (c)  $xy = k$   
 (d)  $y = k \log x$  [b]
51. The following sentence has been divided into parts. One of them contains an error. Select the part that contains the error from the given options.  
 The project was delayed/because of unforeseen issues;/consequence, we had/to adjust our timeline./  
 (a) because of unforeseen issues;  
 (b) consequence, we had  
 (c) to adjust our timeline  
 (d) The project was delayed [b]
52. The following sentence has been split into four segments. Identify the segment that contains an error in phrasal verb.  
 Bindu asked Arya to fill out for / her so that she could / take care of/ her sick father.  
 (a) Bindu asked Arya to fill out for  
 (b) her so that she could  
 (c) her sick father  
 (d) take care of [a]
53. The following sentence has been split into four segments. Identify the segment that contains a grammatical error.

The happy and cheerful girls / is always running towards / the ice cream cart in / front of the garden.

- (a) the ice cream cart in  
 (b) front of the garden  
 (c) is always running towards  
 (d) the happy and cheerful girls [c]

54. The following sentence has been split into four segments. Identify the segment that contains a grammatical error.

Maria was unhappy / because she terribly performed / in the examination / conducted last month.

- (a) Maria was unhappy  
 (b) conducted last month  
 (c) in the examination  
 (d) because she terribly performed [d]

55. Select the option that expresses the given sentence in passive voice.

The stranger had intended the execution of the plot but the boys thwarted it with their subtlety.

- (a) The execution of the plot had been intended by the stranger but it was thwarted by the boys with their subtlety.  
 (b) The execution of the plot could have been intended by the stranger but it was being thwarted by the boys with their subtlety.  
 (c) The execution of the plot was intended by the stranger but it had been thwarted by the boys with their subtlety.  
 (d) The execution of the plot was being intended by the stranger but it has been thwarted by the boys with their subtlety. [a]

56. Select the option that expresses the following sentence in passive voice.

The audience loudly cheered the student leader's speech.

- (a) The student leader's speech was loudly cheered by the audience.  
 (b) There was a loud cheer for the student.  
 (c) They all cheered loudly the leader's speech.  
 (d) The student leader was loudly cheered by the audience. [a]

57. Select the most appropriate word to fill in the blank.

My father was very distant, but I suppose all men of his generation\_\_\_\_\_.

- (a) there  
 (b) was  
 (c) would  
 (d) were [d]

58. Select the most appropriate word to fill in the blank.  
The population \_\_\_\_\_ very fast in the last decade.

(a) had been risen  
(b) rising  
(c) has risen  
(d) rises

[c]

59. Each of the following items in this section consists of a sentence, parts of which have been jumbled. These parts have been labeled as P, Q, R, and S. Given below each sentence are four sequences, namely (a), (b), (c), and (d). You are required to re-arrange the jumbled parts of the sentence and mark your response accordingly.

to find a way to protect(P)/ the world's nature(Q)/ countries are in Montreal(R)/ Delegates from nearly 200(S)

(a) PQRS  
(b) SQRP  
(c) QRPS  
(d) SRPQ

[d]

60. Each of the following items in this section consists of a sentence, parts of which have been jumbled. These parts have been labelled as P, Q, R and S. Given below each sentence are four sequences, namely (a), (b), (c) and (d). You are required to re-arrange the jumbled parts of the sentence and mark your response accordingly.

on the Russian invasion of Ukraine(P)/ In 2022, German users primarily(Q)/ searched Google(R)/ for news and background information(S)

(a) QRSP  
(b) SRPQ  
(c) PQRS  
(d) QRPS

[a]

61. Direction: Read the passage and answer the following questions.

Swami Vivekananda, a monk from India, journeyed to Chicago in 1893, marking a pivotal moment in spiritual history. His mission was to attend the Parliament of the World's Religions, an event that sought to create a global dialogue among various religious faiths. Unbeknownst to him, this voyage would not only introduce the teachings of Vedanta and Yoga to the Western world but also lay the foundation for a deeper understanding and appreciation of Indian spirituality far beyond its borders.

Upon his arrival in Chicago, Vivekananda encountered numerous challenges, including bureaucratic hurdles and financial constraints, which initially hindered his ability to participate in the Parliament. However, his determination and the fortuitous support of a few kind-hearted individuals ensured his attendance at the event. When he addressed the gathering with the words, "Sisters and Brothers of America," he received a standing ovation from thousands of attendees. The cause of his profound impact was his eloquent speech on religious tolerance and universal brotherhood, which resonated deeply with an audience unfamiliar with such inclusivity in spiritual discourse.

The effect of Swami Vivekananda's participation in the Parliament was monumental. His teachings and philosophies gained a significant following in the US and Europe, bridging the cultural and spiritual divide between the East and the West. He established Vedanta Societies, fostering a deeper, cross-cultural exchange of spiritual ideas and practices. Vivekananda's journey to Chicago was not just a personal triumph but a historic event that sowed the seeds of Vedanta and Yoga in the fertile ground of the Western consciousness, enriching it with the profound wisdom of Indian spirituality. His legacy continues to influence seekers of spiritual truth around the world, proving the enduring power of his message of unity and peace.

How was Swami Vivekananda received when he first addressed the gathering at the Parliament of the World's Religions?

(a) With silence  
(b) With skepticism

(c) With a standing ovation

(d) With immediate dismissal

[c]

62. What significant challenges did Swami Vivekananda face upon arriving in Chicago?

(a) Language barriers

(b) Bureaucratic hurdles and financial constraints

(c) Finding a place to stay

(d) All of the above

[b]

63. What event did Swami Vivekananda attend in Chicago?

(a) The International Yoga Day

(b) The Parliament of the World's Religions

(c) The Chicago World's Fair

(d) The United Nations Conference

[b]

64. Which societies did Swami Vivekananda establish to foster spiritual exchange?  
(a) Yoga Societies  
(b) Brotherhood Clubs  
(c) Interfaith Councils  
(d) Vedanta Societies [d]
65. Select the most appropriate ANTONYM of the given word.  
**DISTINCT**  
(a) marked  
(b) prescribed  
(c) noticeable  
(d) vague [d]
66. Find a word that is the ANTONYM OF animosity  
(a) bitterness  
(b) sarcasm  
(c) benevolence  
(d) contamination [c]
67. Choose the most appropriate option and fill in the blank. The Verb form of 'Satisfactory' is \_\_\_\_\_.  
(a) Satisfy  
(b) Satisfying  
(c) Satisfied  
(d) Satisfaction [a]
68. Give a single word that substitutes for the following expression.  
A person who constantly sees the bright side of things  
(a) misogynist  
(b) pessimist  
(c) critic  
(d) optimist [d]
69. Select the option that best gives the meaning of the underlined word.  
He has such an inquisitive mind that he annoys people by his constant questioning.  
(a) Complex  
(b) Mature  
(c) Curious  
(d) Brilliant [c]
70. Fill in the blanks with suitable article.  
My mother is \_\_\_\_\_ honest woman.  
(a) A  
(b) An  
(c) The  
(d) No article required [b]